

Binary Universe Ontology as an Interpretation of Two-Mode Coupling: The Geometric Origin of k_{UAT} as a Speculative Research Program

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Abstract

This technical note interprets the two-mode coupling model of the companion paper [1] within the framework of the Binary Universe Ontology. We identify the analytic sequential mode with Bit 1 (the baryonic sector) and the non-analytic substrate mode with Bit 0 (the non-baryonic sector). We show that the Jaynes–Cummings interaction is a formal realization of causal friction between sectors, that the Lindblad dissipation realizes entropic homeostasis, and that the DMN modulation is the baryonic filter suppressing substrate access. We then present the geometric constants $\Phi_{\text{UAT}} = \pi^2/8$, $k_{\text{UAT}} \approx 10.90677$, and $\kappa_{\text{crit}} \approx 10^{-78}$ as *speculative hypotheses*, not as derived results. The companion paper contains the formal, testable model without these constants. This note is presented as a research program, not as an established result.

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1 Ontological framework

The Binary Universe Ontology posits that observable reality is the interface between two coexisting sectors of a single physical substrate:

- **Bit 1:** the baryonic sector. Localized, finite, deterministic, subject to conservation laws. The domain of sequential causality $A \rightarrow B$ and of the analytic operator $\hat{R}$.
- **Bit 0:** the non-baryonic sector. Non-local, infinite, superposed, quasi-atemporal. The domain of phase coherence and of the instinctive operator $\hat{I}$.

The two sectors are not parallel universes. They interpenetrate in the same spacetime bulk. Their interaction produces what we observe as time, consciousness, and physical law [2, 3].

The Binary Universe Ontology was introduced in [2] and further developed in the series of Zenodo preprints [3, 4, 5]. The present note connects it to a formal dynamical model.

2 Identification with the two-mode model

We identify the formal structures of the companion paper [1] with the ontological framework as follows.

2.1 Analytic mode as Bit 1

The qubit $\mathcal{H}_1 = \mathbb{C}^2$ represents the analytic mode. Its two states are

$$|0\rangle_1 : \text{analytic mode off (Bit 1 disengaged),} \quad (1)$$

$$|1\rangle_1 : \text{analytic mode on (Bit 1 engaged).} \quad (2)$$

The operator

$$\sigma_z = |1\rangle_1 \langle 1| - |0\rangle_1 \langle 0| \quad (3)$$

measures the degree of engagement of the baryonic sector:

$$\langle \sigma_z \rangle = +1 \iff \text{Bit 1 pure,} \quad \langle \sigma_z \rangle = -1 \iff \text{Bit 0 pure.} \quad (4)$$

2.2 Substrate mode as Bit 0

The oscillator $\mathcal{H}_0 = \ell^2(\mathbb{N})$ represents the non-baryonic substrate. Its Fock states $|n\rangle_0$ count the number of substrate quanta. In the Binary Ontology, n_0 is the *degree of activation of the causal substrate*.

The canonical commutator

$$[a, a^\dagger] = 1 \quad (5)$$

encodes the intrinsic indeterminacy of Bit 0: a space of infinite dimension, in contrast to the finite two-level structure of Bit 1.

2.3 Interaction as causal friction

The Jaynes–Cummings term

$$H_{\text{int}} = \hbar g(t) \left(a^\dagger \sigma_- + a \sigma_+ \right) \quad (6)$$

represents the exchange of excitation between sectors:

- $a^\dagger \sigma_-$: an excitation of Bit 1 is transferred to Bit 0 (analytic content is absorbed by the substrate).
- $a \sigma_+$: an excitation of Bit 0 is transferred to Bit 1 (substrate content is manifested analytically).

This is the formal realization of *causal friction*: the baryonic sector cannot evolve without exchanging phase information with the substrate.

2.4 Dissipation as entropic homeostasis

The Lindblad dissipator encodes the thermodynamic interaction between sectors:

- Phase damping (γ_ϕ): loss of analytic coherence, corresponding to the entropic cost of Bit 1.
- Amplitude damping and thermal pumping (γ_a, n_{th}): exchange of substrate quanta, corresponding to the entropic absorption by Bit 0.

The total entropy production satisfies

$$\dot{S}_{\text{total}} = \dot{S}_1 + \dot{S}_0 + \dot{S}_{\text{rad}} \geq 0, \quad (7)$$

while the net baryonic entropy can be stationary:

$$\dot{S}_{1,\text{net}} = \dot{S}_1 + \dot{S}_{1 \leftarrow 0} = 0. \quad (8)$$

This is the formal realization of the causal homeostasis postulated in the Binary Ontology.

2.5 Mixing angle as interface threshold

The mixing angle

$$\theta_n = \frac{1}{2} \arctan \left(\frac{2g\sqrt{n+1}}{\delta} \right) \quad (9)$$

measures the admixture between sectors. The regimes correspond to states of consciousness:

θ_n	Sector dominance	Phenomenology
0	Bit 1 pure	Ordinary waking, analytic dominance
$\pi/8$	Threshold	Onset of substrate access
$\pi/4$	Maximal mixing	Flow, dual consciousness (asymptotic)
$\pi/2$	Bit 0 pure	Deep sleep, analytic off

The companion paper [1] showed that the transition between analytical dominance and substrate dominance is a curve in the (α, γ_ϕ) plane, not a single threshold. The Binary Ontology accommodates this: the interface between sectors is modulated by both coupling and dissipation, not by coupling alone.

2.6 DMN as baryonic filter

The Default Mode Network is identified with the baryonic filter that suppresses access to Bit 0. Its activity $A_{\text{DMN}}(t)$ modulates the coupling:

$$g(t) = g_0 \eta(A_{\text{DMN}}(t)), \quad \eta(A) = \frac{1}{1 + e^{\lambda(A - A_{\text{th}})}} \quad (10)$$

When $A_{\text{DMN}} \rightarrow 0$, the suppression vanishes and Bit 0 becomes accessible. This is a formal statement of the Binary Ontology's claim that the analytic sector actively suppresses the substrate sector.

3 Endogenous vs exogenous access as Bit 0 modes

The model distinguishes two routes to Bit 0 access, corresponding to different regimes of the phase diagram.

3.1 Endogenous access

$$g \text{ moderate, } \gamma_\phi \text{ low, } \langle n_0 \rangle_{ss} \approx 0.2. \quad (11)$$

The coupling increases without a corresponding increase in noise. The phase distribution is concentrated and the channel capacity

$$C = B \log_2(1 + \kappa), \quad \kappa = g/(\gamma_\phi + \gamma_a), \quad (12)$$

grows logarithmically. This is the regime of high-fidelity information transfer from Bit 0 to Bit 1. In the companion paper, this corresponds to the endogenous case with $\langle \sigma_z \rangle_{ss} > 0$ and moderate substrate excitation.

3.2 Exogenous access

$$g \text{ high, } \gamma_\phi \text{ high, } \langle n_0 \rangle_{ss} \approx 0.4. \quad (13)$$

The coupling increases but dissipation saturates the substrate. The subject experiences an apparent infinity of information without retaining any of it. In the companion paper, this corresponds to the exogenous case with $\langle \sigma_z \rangle_{ss} < 0$ and maximal substrate excitation.

3.3 Implications for consciousness studies

The formal distinction between the two regimes provides a mathematically precise language for phenomena reported in contemplative neuroscience and psychedelic research. Voluntary access to altered states (as in advanced meditation, or in isolated cases reported in the fMRI literature) corresponds to the endogenous regime. Pharmacologically forced access (classical psychedelics) corresponds to the exogenous regime. The two are quantitatively distinguishable in the steady-state substrate excitation $\langle n_0 \rangle_{ss}$.

4 Speculative extension: geometric constants

4.1 Status of the constants

The Binary Ontology introduces three dimensionless constants:

$$\Phi_{\text{UAT}} = \frac{\pi^2}{8} \approx 1.2337, \quad k_{\text{UAT}} = e^{g\tau} \approx 10.90677, \quad \kappa_{\text{crit}} \approx 10^{-78}. \quad (14)$$

These constants are not derived in the present work. They are presented as speculative proposals. The formal model of the companion paper [1] does not use them. If future data fit them, they may acquire empirical status; until then, they remain hypotheses.

4.2 Conjectured identifications

If the Binary Ontology is correct, the following identifications are conjectured:

- $\Phi_{\text{UAT}} = \pi^2/8$: the geometric efficiency of the substrate filter. It would appear as the asymptotic value of the coupling modulation in the deep endogenous regime.
- $k_{\text{UAT}} \approx 10.90677$: the critical gain for the transition to the substrate-dominated regime, in units of the dissipative rate.
- $\kappa_{\text{crit}} \approx 10^{-78}$: the saturation threshold of the substrate, below which Bit 0 remains coherent.

These identifications are falsifiable: they predict that the fitted values of g_0 , λ , and A_{th} in the companion model should scale with these constants. If they do not, the conjecture is refuted.

4.3 Relation to the anti-frequency formalism

The anti-frequency $\lambda \equiv -1/f$ was introduced in [4] as the phase response of the Bit 0 substrate. In the two-mode formalism, this corresponds to the phase of the oscillator mode a . The conjugate quadratures (X, P) of the oscillator, defined by $a = (X + iP)/\sqrt{2}$, play the role of the "phase" and "amplitude" of the substrate response. The variance of the P -quadrature is the formal analog of the anti-frequency intensity.

This identification is currently qualitative. A quantitative formulation would require mapping the anti-frequency spectrum to the oscillator spectrum, which is beyond the scope of this note.

5 Falsifiability and research program

5.1 The two tracks are independent

The companion paper (Track A) treats g_0 , λ , A_{th} , γ_ϕ , γ_a , n_{th} , and τ_{DMN} as free parameters to be fitted to data. This note (Track B) proposes that some of these parameters may be fixed by geometric constants. The two tracks are independent:

- Track A: a formal model of neural coupling, testable now.
- Track B: a speculative interpretation in terms of a binary ontology, testable in the future.

If future data fit the constants, the Binary Ontology gains empirical support. If they do not, the formal model of Track A remains valid.

5.2 Specific future tests

1. **High-resolution fMRI of trained practitioners.** Fit the endogenous case of Track A to data from subjects trained in voluntary state transitions. Report the fitted g_0 and γ_ϕ . If g_0/γ_ϕ converges to a universal value across subjects, compare to $k_{\text{UAT}} \approx 10.9$.
2. **Pharmacological contrast.** Fit the exogenous case to data from psilocybin or LSD studies. Report the fitted values of the same parameters. Compare with the endogenous case at equal $\mathcal{R}$.
3. **Gravitational-wave background searches.** The 84.44 Hz monochromatic peak was proposed in [2] as a signature of the substrate's Compton frequency. Current and future interferometers (LIGO O4/O5, Einstein Telescope, Cosmic Explorer) can test this prediction.

4. **Atomic-clock tests.** The temporal retardation of ~ 400 ns/day proposed in [2] for a resonant plasma chamber is at the detection threshold of current optical lattice clocks. A null result would constrain the coupling.

6 Conclusion

We have shown that the two-mode coupling model of the companion paper admits a natural interpretation within the Binary Universe Ontology:

1. Bit 1 is the analytic mode (qubit).
2. Bit 0 is the substrate mode (oscillator).
3. The Jaynes–Cummings interaction is causal friction.
4. The Lindblad dissipation is entropic homeostasis.
5. The mixing angle θ_n measures the interface between sectors.
6. The DMN is the baryonic filter suppressing substrate access.
7. The distinction between endogenous and exogenous access is formalized by the steady-state substrate excitation.

The formal content of the model is testable with current EEG and fMRI techniques. The speculative extension to geometric constants is presented as a research program, not as an established result. The two tracks are kept separate to preserve the scientific integrity of both.

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